

Class Exercise 1: Diabetes Prediction Using Machine Learning Classification

Course: Intelligent AI and Data Engineering

Module: Machine Learning - Classification

Exercise Type: Individual Practical Lab

Difficulty: Beginner

Estimated Time: 2–3 Hours

Dataset: `diabetes.csv`

Objective

In this exercise, you will build your **first Machine Learning Classification model** using the Diabetes dataset.

Your goal is to predict whether a patient has diabetes based on medical information.

You will follow the complete Machine Learning workflow from data exploration to model evaluation.

Learning Outcomes

After completing this exercise, students should be able to:

- Understand a binary classification problem
- Explore and understand a dataset
- Clean missing or invalid values
- Perform Exploratory Data Analysis (EDA)
- Prepare data for Machine Learning
- Train multiple classification models
- Compare different algorithms

- Evaluate classification models
 - Save the best model for future use
-

Business Problem

Hospitals collect medical information about patients during routine checkups.

Doctors would like to identify patients who are likely to have diabetes before symptoms become severe.

By building a Machine Learning model, hospitals can:

- Detect diabetes early
- Assist doctors during diagnosis
- Improve patient care
- Reduce manual screening time

Your task is to build a model that predicts whether a patient has diabetes.

Dataset Description

The dataset contains **768 patient records**.

Each row represents one patient.

Target Variable

Outcome

Value	Meaning
1	Patient has diabetes
0	Patient does not have diabetes

Dataset Features

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Blood glucose concentration
BloodPressure	Blood pressure
SkinThickness	Skin fold thickness
Insulin	Insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes hereditary score
Age	Patient age
Outcome	Target variable

Exercise 1: Load the Dataset

Tasks

- Import Pandas
- Import NumPy
- Load the dataset
- Display the first five rows
- Display the last five rows
- Display dataset shape

Exercise 2: Explore the Dataset

Answer the following questions.

1. How many rows are there?
2. How many columns?
3. What are the feature names?
4. Which column is the target variable?
5. What are the data types?
6. Are there missing values?
7. Display statistical summary.

Use

```
df.info()  
  
df.describe()  
  
df.isnull().sum()
```

Exercise 3: Data Cleaning

Part A

Check duplicate rows.

If duplicates exist,

remove them.

Part B

Some columns contain **0 values** that are medically impossible.

Check the following columns:

- Glucose
- BloodPressure

- SkinThickness
- Insulin
- BMI

Count the number of zero values.

Part C

Replace all impossible zero values with **NaN**.

Part D

Replace missing values using the **median** of each column.

Exercise 4: Exploratory Data Analysis (EDA)

Create the following visualizations.

Chart 1

Diabetes Distribution

Questions

- Is the dataset balanced?
 - Which class appears more often?
-

Chart 2

Age Distribution

Questions

- Which age group appears most?
 - Is the distribution symmetric?
-

Chart 3

Glucose Distribution

Questions

- Are glucose values normally distributed?
 - Are there outliers?
-

Chart 4

BMI Distribution

Questions

- Are most patients overweight?
 - Are there extreme values?
-

Chart 5

Correlation Heatmap

Questions

- Which feature has the strongest correlation with Outcome?
 - Which two features are highly correlated?
-

Chart 6

Boxplot

Glucose vs Outcome

Question

Do diabetic patients generally have higher glucose levels?

Exercise 5: Prepare the Data

Separate the dataset into

Features

```
X
```

and

Target

```
y
```

Target column

```
Outcome
```

Exercise 6: Split the Dataset

Split the data into

- 80% Training
- 20% Testing

Use

```
random_state=42
```

Exercise 7: Feature Scaling

Scale the features using

```
StandardScaler
```

Questions

1. Why is feature scaling important?
2. Why do we use

```
fit_transform()
```

only on the training data?

3. Why do we use

```
transform()
```

on the testing data?

Exercise 8: Train Multiple Models

Train the following classifiers.

Model 1

Logistic Regression

Model 2

Decision Tree

Model 3

Random Forest

Model 4

Support Vector Machine (SVM)

Model 5

K-Nearest Neighbors (KNN)

Exercise 9: Model Evaluation

Evaluate every model using

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- Classification Report

Create a comparison table.

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression				
Decision Tree				
Random Forest				
SVM				
KNN				

Exercise 10: Choose the Best Model

Answer the following questions.

1. Which model achieved the highest Accuracy?
2. Which model achieved the highest Recall?
3. Which model achieved the highest F1 Score?
4. Which model would you recommend?

Explain your answer.

Exercise 11: Save the Best Model

Save the best-performing model.

Requirements

- Save the model as

```
models/best_model.pkl
```

using Joblib.

- Save the feature names as

```
models/feature_info.json
```

Exercise 12: Interpretation Questions

Answer the following.

Question 1

What is Classification?

Question 2

Why is this a binary classification problem?

Question 3

Explain

- Accuracy
 - Precision
 - Recall
 - F1 Score
-

Question 4

What is a False Positive?

Give a real-life example.

Question 5

What is a False Negative?

Give a real-life example.

Question 6

Which error is more dangerous in diabetes prediction?

Why?

Question 7

Why should we save the trained model?

Question 8

Why should we save the feature metadata?